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Deepfake Detection Using CNN and Capsule Networks

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DECLARATION

I hereby certify that the work presented in the project report entitled "Deepfake Detection Using CNN and Capsule Networks" in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, submitted to the Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Ms. Shikha Tuteja. The matter presented in this project report has not been submitted by me elsewhere for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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CONTENTS

ABSTRACT

The rapid advancement of deep learning technologies has enabled the creation of highly realistic synthetic media, commonly known as deepfakes. These AI-generated videos and images pose significant risks to digital security, personal privacy, and the integrity of public information. While deepfakes find legitimate applications in entertainment and media production, their potential for misuse — including identity fraud, disinformation campaigns, and political manipulation — has emerged as a pressing global concern.

Existing detection approaches predominantly rely on Convolutional Neural Networks (CNNs), which, while powerful in feature extraction, are limited in their ability to preserve spatial hierarchies and part-whole relationships across facial components. This inherent limitation reduces their effectiveness in detecting structurally subtle manipulations. Capsule Networks (CapsNets) have been proposed as an alternative due to their capacity to retain spatial configuration and orientation information; however, standalone CapsNet models face scalability challenges and require further validation on large-scale datasets.

This report presents a hybrid deep learning framework that integrates CNNs with Capsule Networks for robust deepfake detection. The CNN component is responsible for extracting multi-scale visual features, including edges, textures, and manipulation artifacts. These features are subsequently fed into the Capsule Network, which applies dynamic routing to encode spatial relationships between facial components. The resulting architecture captures both low-level visual anomalies and high-level structural inconsistencies inherent in deepfake media.

The model was trained and evaluated on two widely-used benchmark datasets: FaceForensics++ and Celeb-DF. Preprocessing steps included frame sampling, face extraction using MTCNN, image normalization, and data augmentation techniques such as horizontal flipping, rotation, and brightness variation. The proposed hybrid model achieved 94.3% accuracy and an F1-score of 0.93, outperforming standalone CNN (89.2% accuracy) and CapsNet (91.0% accuracy) models. Cross-dataset evaluation demonstrated a generalization accuracy of 91.5% on unseen data.

The experimental results confirm that the synergistic combination of CNN and Capsule Networks provides a more discriminative and generalizable framework for deepfake detection. This work contributes a reliable baseline for future research in automated synthetic media forensics.

Keywords: Deepfake Detection, Convolutional Neural Network, Capsule Networks, Synthetic Media, Dynamic Routing, FaceForensics++, Celeb-DF, Artificial Intelligence.

1. Introduction

The proliferation of deep learning technologies over the past decade has brought with it a growing class of synthetic media known as deepfakes. These are digitally manipulated images and videos in which a person's face, voice, or mannerisms have been replaced or altered using generative models, most notably Generative Adversarial Networks (GANs) and autoencoders. While such technology has enabled creative applications in film production, virtual reality, and digital arts, its misuse has raised serious ethical, legal, and security concerns worldwide.

Deepfake videos have been used to fabricate statements by political figures, produce non-consensual intimate imagery, perpetrate financial fraud through identity deception, and amplify disinformation across social media platforms. The increasing photorealism of modern deepfakes makes them extremely difficult to detect with the naked eye, even for trained observers. As deepfake generation technology continues to improve, the gap between human detection capability and machine-generated realism widens further.

Early detection approaches relied on manually engineered features — such as blinking anomalies, lighting inconsistencies, and compression artifacts — to identify manipulated content. However, the sophistication of contemporary deepfake generation methods has rendered these handcrafted approaches largely ineffective. The research community has since turned to automated deep learning-based detection, with Convolutional Neural Networks (CNNs) emerging as the dominant paradigm.

CNNs have demonstrated remarkable capability in extracting hierarchical visual representations, enabling the detection of texture anomalies, blending artifacts, and pixel-level distortions associated with facial manipulation. Popular architectures such as XceptionNet, EfficientNet, and ResNet have been adapted for this task with considerable success. Nevertheless, CNNs are not without limitations. The pooling operations inherent to CNN architectures result in the loss of spatial information, limiting the model's ability to reason about the structural relationships between different facial components.

To address this limitation, Capsule Networks (CapsNets) were proposed as an alternative architecture that encodes not only the presence of features but also their spatial orientation and relationships. CapsNets represent features as vectors (capsules) rather than scalar activations, and employ a dynamic routing mechanism to establish part-whole relationships between lower and higher-level entities. This makes them particularly well-suited for detecting subtle structural inconsistencies in manipulated facial images.

Recent research has explored the combination of CNNs and Capsule Networks to leverage the complementary strengths of both architectures. CNNs provide efficient, multi-scale feature extraction, while CapsNets enhance spatial reasoning and generalization. This hybrid approach has demonstrated improved robustness across diverse datasets and manipulation types.

This report presents the design, implementation, and evaluation of a hybrid CNN-CapsNet model for deepfake detection. The model is trained and tested on the FaceForensics++ and Celeb-DF benchmark datasets, and its performance is compared against standalone CNN and CapsNet baselines. The following sections detail the literature landscape, the proposed methodology, the system architecture, the experimental setup, and the results obtained.

1.1 Problem Statement

The central challenge addressed in this work is the development of an automated deepfake detection system capable of overcoming the limitations of existing approaches. Specifically, CNN-based models lose critical spatial information during feature aggregation, while standalone CapsNet models lack the scalability required for large-scale deployment. There is a need for a hybrid architecture that combines the feature extraction power of CNNs with the spatial relationship modeling of Capsule Networks to deliver accurate, robust, and generalizable deepfake detection.

1.2 Objectives

1. Design and implement a hybrid deep learning model integrating CNN and Capsule Networks for binary deepfake classification.
2. Develop a comprehensive preprocessing pipeline for video data, including face extraction, normalization, and data augmentation.
3. Train and evaluate the model on standard benchmarks: FaceForensics++ and Celeb-DF datasets.

4. Achieve superior performance in terms of accuracy, precision, recall, and F1-score compared to standalone models.
5. Assess the model's cross-dataset generalization capability to validate real-world applicability.

2. Literature Review

The field of deepfake detection has evolved significantly from early handcrafted approaches to sophisticated deep learning architectures. This section reviews the major phases of this evolution and highlights the comparative strengths and limitations of each approach.

2.1 Traditional and Handcrafted Feature-Based Methods

The earliest deepfake detection methods relied on manually engineered feature extractors designed to capture specific visual artifacts introduced during the synthesis process. These methods analysed indicators such as facial landmark inconsistencies, unnatural eye blinking patterns, irregular head pose dynamics, and luminance discontinuities at facial boundaries. Forensic-based approaches also examined compression artifacts and statistical noise patterns in the frequency domain.

While these methods were computationally efficient and interpretable, they proved brittle in the face of advancing generation techniques. Modern GAN-based deepfakes have effectively eliminated many of the classic artifacts, rendering handcrafted detectors largely obsolete in real-world scenarios where the generation method is unknown.

2.2 CNN-Based Detection Methods

The advent of deep learning, particularly CNNs, transformed the detection landscape. Architectures such as XceptionNet, EfficientNet, and ResNet were adapted for deepfake classification by learning to detect subtle visual anomalies that are difficult to perceive manually. Rossler et al. (2019) introduced FaceForensics++, a landmark benchmark dataset alongside an XceptionNet-based baseline that demonstrated the power of CNNs for this task.

CNN-based models excel at extracting hierarchical representations — from low-level edges and textures to high-level semantic features — through successive convolutional and pooling layers. However, the pooling operations that enable spatial invariance also discard precise spatial information, limiting the model's capacity to reason about the relational structure of facial components. Furthermore, CNN models trained on specific datasets often struggle to generalize to new manipulation types or unseen datasets.

2.3 Capsule Network-Based Approaches

Hinton et al.'s concept of Capsule Networks addressed a fundamental shortcoming of CNNs by representing features as vectors (capsules) encoding both the presence and the pose of entities. The dynamic routing mechanism enables CapsNets to learn part-whole relationships between features, making them uniquely capable of detecting structural inconsistencies across facial regions. Mehra et al. (2021) demonstrated the applicability of CapsNets to deepfake detection, showing competitive performance with fewer parameters than deep CNN models.

Despite their theoretical advantages, standalone CapsNets face practical challenges including high training complexity, slower convergence, and limited scalability to large datasets. They also tend to underperform CNN architectures in tasks requiring rich, multi-scale feature extraction.

2.4 Hybrid CNN + Capsule Network Methods

Recognizing the complementary nature of CNNs and CapsNets, recent research has focused on hybrid architectures that use CNNs for efficient feature extraction and CapsNets for spatial reasoning. Ishrak et al. (2024) proposed an explainable deepfake detection framework combining both architectures, demonstrating improved accuracy and interpretability. Saikia et al. (2022) explored CNN-LSTM hybrids for temporal deepfake detection in video streams.

The comparative analysis of detection approaches is summarized in Table 1 below.

Table 1: Comparative Analysis of Detection Approaches

Approach	Key Strength	Primary Limitation

Handcrafted Features	Interpretable, low cost	Fails on modern deepfakes
CNN-Based Models	High accuracy, scalable	Loss of spatial info via pooling
Capsule Networks	Preserves spatial hierarchies	High training complexity
CNN + LSTM/RNN	Temporal feature capture	Very high compute cost
Proposed: CNN + CapsNet	94.3% accuracy; generalizable	Needs real-time optimization

3. Proposed Methodology

The proposed methodology consists of five principal stages: data acquisition and preprocessing, CNN-based feature extraction, Capsule Network encoding, classification, and model evaluation. Each stage is described in detail below.

3.1 Dataset Description

Two publicly available, widely-used deepfake detection benchmarks were employed in this study:

- FaceForensics++ (FF++): A large-scale dataset containing over 1,000 original video sequences and their corresponding manipulated counterparts, generated using four manipulation methods — DeepFakes, Face2Face, FaceSwap, and NeuralTextures. Videos are available at multiple compression levels (RAW, HQ, LQ).
- Celeb-DF: A high-quality deepfake dataset comprising 590 original celebrity interview videos and 5,639 synthesized deepfake videos. Celeb-DF was designed to address the relatively low visual quality of earlier datasets and presents a more challenging detection scenario.

3.2 Data Preprocessing

Raw video data from both datasets underwent a multi-step preprocessing pipeline before being fed into the model:

1. Frame Sampling: Individual frames were extracted from each video at a fixed sampling frequency to create an image-level dataset.
2. Face Detection and Extraction: The MTCNN (Multi-task Cascaded Convolutional Networks) face detector was applied to each frame to localize and crop the facial region of interest.

3. Resizing and Normalization: Extracted face crops were resized to a uniform resolution and pixel values were normalized to the range [0, 1] to standardize input distribution.
4. Data Augmentation: To reduce overfitting and improve generalization, the training set was augmented using random horizontal flips, rotations (±15°), and brightness and contrast adjustments.

3.3 CNN Feature Extraction

The CNN component of the hybrid architecture serves as the primary feature extractor. It processes the normalized facial image through a series of convolutional layers, each followed by batch normalization and ReLU activation. Hierarchical feature maps are produced that encode progressively more abstract representations — from low-level edges and texture patterns to high-level semantic features and manipulation artifacts. The final convolutional output is passed to the Capsule Network as a structured feature representation.

3.4 Capsule Network Architecture

The Capsule Network module receives the feature maps from the CNN and processes them through two principal layers:

- Primary Capsule Layer: Convolutional feature maps are transformed into vector-valued capsules. Each capsule encodes the presence and spatial properties of a specific facial feature.
- Higher-Level Capsule Layer (DigitCaps): Dynamic routing by agreement is applied between primary and higher-level capsules. This mechanism ensures that each higher-level capsule receives input from lower-level capsules that are in spatial agreement with it, preserving part-whole relationships and hierarchical structure.

The length (norm) of each output capsule vector represents the probability that the corresponding entity is present in the input. A fully connected layer then uses these capsule outputs to produce the final binary classification — Real or Fake.

3.5 Training Configuration

The model was trained using a supervised learning approach with cross-entropy loss. The Adam optimiser was used with an initial learning rate of 1e-4 and weight decay regularisation.

Training was performed over 50 epochs with a batch size of 32. Early stopping with a patience of 10 epochs was applied to prevent overfitting. The dataset was split into training (70%), validation (15%), and testing (15%) subsets.

4. Tools and Technologies

The implementation of the proposed system leveraged a range of open-source tools and frameworks; each selected for its suitability to the task:

Tools / Framework	Purpose
Python 3.x	Primary programming language for model development and experimentation
TensorFlow / Keras	Deep learning framework for building, training, and evaluating CNN and CapsNet models
PyTorch	Alternative framework used for experimental CapsNet implementations
OpenCV	Video processing, frame extraction, and image manipulation
MTCNN / Dlib	Multi-task CNN-based face detection and facial landmark localization
Scikit-learn	Model evaluation: accuracy, precision, recall, F1-score, confusion matrix
Matplotlib / Seaborn	Visualization of training curves, accuracy charts, and confusion matrices
FaceForensics++	Primary deepfake benchmark dataset for training and testing
Celeb-DF	Secondary dataset used for cross-dataset generalization evaluation

Table 2: Tools and Technologies Used

5. Results and Analysis

The proposed hybrid CNN-CapsNet model was evaluated against standalone CNN and Capsule Network baselines on both the FaceForensics++ and Celeb-DF datasets. Performance was assessed using four standard classification metrics: Accuracy, Precision, Recall, and F1-Score.

5.1 Performance Comparison

Table 3 summarises the quantitative performance of all three models on the primary test set.

Table 3: Model Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score
CNN Only	89.2	0.89	0.88	0.88
Capsule Network Only	91.0	0.90	0.91	0.89
Hybrid CNN + CapsNet (Proposed)	94.3	0.94	0.93	0.93
Cross-Dataset (Celeb-DF)	91.5	0.91	0.92	0.91

5.2 Analysis of Results

The hybrid model demonstrated a statistically significant improvement over both individual baselines. The 5.1 percentage point gain in accuracy over the standalone CNN model and the 3.3 percentage point gain over the standalone CapsNet model reflect the synergistic benefit of combining complementary architectural strengths.

The confusion matrix analysis revealed a True Positive Rate (TPR) of 92.0% (920 real samples correctly classified) and a True Negative Rate (TNR) of 93.0% (930 fake samples correctly detected), with only 150 total misclassifications out of 2,000 test samples. This balanced distribution of errors indicates that the model does not exhibit strong class-specific bias.

Feature importance analysis identified Texture (0.22) and Facial Geometry (0.20) as the most discriminative features for deepfake classification, followed by Edges (0.18), Artifacts (0.15), and Frequency-domain patterns (0.12). This finding aligns with the theoretical expectation that deepfake manipulations introduce characteristic irregularities in facial texture and structural geometry.

5.3 Cross-Dataset Generalization

The generalization evaluation on the Celeb-DF dataset — which was not used during training — yielded an accuracy of 91.5%. This result is particularly significant because it demonstrates that the hybrid model can adapt to a substantially different deepfake generation methodology without any fine-tuning. The Capsule Network's capacity to model spatial relationships appears to contribute to this transferability, as structural manipulation patterns tend to be consistent across generation methods even when visual appearance varies.

6. Conclusion and Future Scope

6.1 Conclusion

This report has presented a comprehensive study of deepfake detection using a hybrid deep learning architecture that integrates Convolutional Neural Networks with Capsule Networks. The proposed model capitalizes on the complementary strengths of both architectures: CNNs provide powerful, multi-scale feature extraction, while Capsule Networks encode spatial relationships and part-whole hierarchies through dynamic routing.

Experimental evaluation on the FaceForensics++ and Celeb-DF datasets confirmed that the hybrid model achieves 94.3% accuracy and 0.93 F1-score — superior to both standalone CNN (89.2%) and CapsNet (91.0%) baselines. Cross-dataset testing with 91.5% accuracy demonstrated strong generalization capability, a critical requirement for real-world deployment. The confusion matrix and feature importance analysis further validated the model's discriminative power, particularly in detecting texture and facial geometry inconsistencies.

These results establish the hybrid CNN-CapsNet framework as a reliable and robust approach for automated deepfake forensics, providing a strong baseline for continued research in this domain.

6.2 Future Scope

Despite the promising results, several avenues for future improvement remain:

- Lightweight Architecture Design: The current model demands substantial computational resources, limiting its deployment in real-time and resource-constrained environments. Future work should explore knowledge distillation, model pruning, and neural architecture search to produce efficient variants suitable for mobile and edge devices.
- Multimodal Detection: Current detection relies exclusively on visual features. Incorporating audio analysis — particularly lip synchronization and voice consistency — would strengthen detection robustness against audio-visual deepfakes.
- Adversarial Robustness: The model's resilience to adversarial perturbations and adversarially crafted deepfakes designed to evade detection systems warrants investigation. Adversarial training and certified defences represent promising directions.
- Explainability and Interpretability: Integrating Gradient-weighted Class Activation Mapping (Grad-CAM) or attention visualization mechanisms would improve transparency, making the model more suitable for digital forensics applications where prediction rationale must be understood.
- Real-Time Deployment: Optimization for live-stream analysis and social media monitoring pipelines would extend the practical utility of the proposed system significantly.
- Continual Learning: As deepfake generation methods continue to evolve, detection models must be updated continuously. Exploring continual learning strategies that enable incremental adaptation without catastrophic forgetting is an important long-term objective.

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